Vocabulary: Causal Inference and ML — Robust Estimation of Treatment Heterogeneity

Kristian Boroz

May 11, 2026

Potential Outcomes and Identification

- **Potential Outcomes** $Y_i(0)$, $Y_i(1)$: the outcomes unit i *would* experience under control and treatment, respectively. Only one is ever observed. *The fundamental problem of causal inference: we never see both at once.*

- **Conditional Average Treatment Effect (CATE)**:

$$\tau(x) = \mathbb{E}[Y_i(1) - Y_i(0) \mid X_i = x].$$

  The expected treatment effect for individuals with covariates $X = x$. *Goal: move beyond a single population average to a function of characteristics.*

- **Average Treatment Effect (ATE)**:

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)].$$

  The population-level average; a single number, not a function of x. *The CATE reduces to the ATE when $\tau(x)$ is constant.*

- **Treatment Heterogeneity (HTE)**: the phenomenon where the causal impact of an intervention varies across individuals or subgroups. *Example: a subsidy increases investment for small firms but not large ones.*

- **Unconfoundedness** (Conditional Ignorability):

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i.$$

  Treatment assignment is "as good as random" once we condition on observed covariates X_i. *The key identifying assumption for observational CATE estimation.*

- **Overlap** (Positivity): $0 < e(x) < 1$ for all x, where $e(x) = \mathbb{P}(W_i = 1 \mid X_i = x)$ is the propensity score. *Ensures that both treated and control units exist in every region of covariate space.*

- **SUTVA** (Stable Unit Treatment Value Assumption): the potential outcomes of unit i do not depend on the treatment assigned to other units, and there is only one version of treatment. *Rules out spillovers and interference between units.*

Bias Sources

- **Confounding Bias**: distortion in a treatment effect estimate caused by variables that affect both treatment assignment and the outcome. In observational data:

$$\mathbb{E}[Y_i \mid W_i = 1] - \mathbb{E}[Y_i \mid W_i = 0] \neq \tau$$

  because treated and control units differ systematically *before* treatment. *Addressed by conditioning on X_i under unconfoundedness.*

- **Selection Bias**: a form of confounding where units *select* into treatment based on characteristics correlated with the outcome. *Example: healthier patients choosing surgery inflates estimated surgical benefit.*

- **Regularization Bias**: distortion in CATE estimates when an ML model's regularization (e.g. shrinkage, depth limits) affects the treatment and control functions differently — typically because $n_1 \neq n_0$. *Can "hallucinate" heterogeneity even when the true effect is constant.*

- **Omitted Variable Bias (OVB)**: bias from a confounder that is correlated with both W_i and Y_i but is not included in X_i. *Unconfoundedness fails; OVB cannot be removed by better ML models alone.*

Nuisance Functions and Propensity

- **Propensity Score** $e(x)$:

$$e(x) = \mathbb{P}(W_i = 1 \mid X_i = x) = \mathbb{E}[W_i \mid X_i = x].$$

  The probability of receiving treatment given covariates. *Used in inverse-probability weighting, X-learner, and Robinson's transformation.*

- **Outcome Regression** $\mu^{(w)}(x)$:

$$\mu^{(w)}(x) = \mathbb{E}[Y_i \mid X_i = x, W_i = w], \quad w \in \{0, 1\}.$$

  The conditional mean outcome for each treatment arm. *Central building block of S-, T-, and X-learners.*

- **Marginal Outcome Mean** $m(x)$:

$$m(x) = \mathbb{E}[Y_i \mid X_i = x].$$

  The unconditional (on W) conditional mean of Y. *One of the two nuisance functions estimated in Robinson's transformation.*

- **Nuisance Component**: any function that must be estimated to identify the target parameter but is not itself of substantive interest (e.g. $m(x)$ and $e(x)$ in Robinson's estimator). *Must be estimated well; errors propagate into the treatment effect estimate.*

Meta-Learners

- **Meta-Learner**: a strategy that combines off-the-shelf ML predictors to estimate the CATE. The "meta" refers to how base learners are combined, not to a specific algorithm.

- **S-Learner** (Single learner): fits one model $\hat{\mu}(x, w)$ with W included as a feature, then estimates:

$$\hat{\tau}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0).$$

  Risk: regularization may shrink the coefficient on W, biasing $\hat{\tau}$ toward zero.

- **T-Learner** (Two-model learner): fits separate models $\hat{\mu}^{(0)}(x)$ and $\hat{\mu}^{(1)}(x)$ on the control and treated samples, then:

$$\hat{\tau}(x) = \hat{\mu}^{(1)}(x) - \hat{\mu}^{(0)}(x).$$

  Risk: different effective regularization per group can hallucinate heterogeneity.

- **X-Learner** (Cross learner, Künzel et al. 2019): reduces regularization bias by imputing *pseudo-effects* for each unit and weighting by the propensity score:

$$D_i^{(1)} = Y_i - \hat{\mu}^{(0)}(X_i), \qquad D_i^{(0)} = \hat{\mu}^{(1)}(X_i) - Y_i,$$

$$\hat{\tau}(x) = \hat{e}(x)\,\hat{\tau}^{(1)}(x) + \big(1 - \hat{e}(x)\big)\hat{\tau}^{(0)}(x).$$

  Particularly useful when treatment and control groups are very different in size.

- **Pseudo-Effect** $D_i^{(w)}$: an imputed individual-level treatment effect estimate constructed by comparing an observed outcome to a model-predicted counterfactual. *Used as the outcome in the second stage of the X-learner.*

Robinson's Transformation and Double Robustness

- **Partial Linear Model**: a semiparametric model of the form

$$Y_i = \tau W_i + \mu^{(0)}(X_i) + \varepsilon_i,$$

  where τ is the constant treatment effect and $\mu^{(0)}(x)$ is a flexible nuisance baseline. *Robinson (1988) showed how to estimate τ without specifying $\mu^{(0)}$.*

- **Robinson's Transformation** (Residual-on-Residual Regression): subtract conditional means to eliminate the nuisance:

$$\underbrace{Y_i - m(X_i)}_{\tilde{Y}_i} = \tau\,\underbrace{\big(W_i - e(X_i)\big)}_{\tilde{W}_i} + \varepsilon_i,$$

  then estimate by OLS:

$$\hat{\tau} = \frac{\sum_i \tilde{Y}_i\,\tilde{W}_i}{\sum_i \tilde{W}_i^2}.$$

  Nuisance functions $\hat{m}$ and $\hat{e}$ can be estimated with any ML method.

- **Double Robustness**: the property that an estimator remains consistent if *at least one* of the two nuisance models ($\hat{m}$ or $\hat{e}$) is correctly specified. *Robinson's estimator and causal forests both enjoy this property.*

- **Local Robustness** (Neyman Orthogonality): the first-order bias from nuisance estimation errors cancels out in the estimating equation. *Small errors in $\hat{m}$ or $\hat{e}$ have only second-order effects on $\hat{\tau}$.*

- **Semiparametric Efficiency**: an estimator achieves the lowest possible asymptotic variance among all regular estimators in the semiparametric model. *Robinson's estimator attains this bound.*

Causal Forests

- **Adaptive Weights** $\alpha_i(x)$: forest-induced similarity weights assigned to each training observation when predicting at x:

$$\alpha_i(x) = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}[x \text{ and } X_i \text{ fall in the same leaf in tree } b].$$

  A forest acts as a data-adaptive kernel: units "close" to x in relevant dimensions receive higher weight.

- **Causal Forest**: an ensemble of honest causal trees that estimates the CATE via a localized Robinson regression:

$$\hat{\tau}(x) = \frac{\sum_i \alpha_i(x)\,(Y_i - \hat{m}(X_i))(W_i - \hat{e}(X_i))}{\sum_i \alpha_i(x)\,(W_i - \hat{e}(X_i))^2}.$$

  Confounding-robust by construction; handles high-dimensional X without explicit modeling.

- **Causal Splitting Criterion**: instead of maximizing outcome variance (as in regression trees), causal forests select splits that maximize heterogeneity in *treatment effects*:

$$\max_{j,\,s}\; n_L n_R \big(\hat{\tau}_L - \hat{\tau}_R\big)^2.$$

  Focuses the tree structure on drivers of effect heterogeneity, not drivers of the outcome level.

- **Honest Estimation**: the sample is split so that one part determines the tree structure (splits) and a separate part estimates treatment effects within leaves. *Prevents overfitting and enables valid confidence intervals.*

- **Localized Robinson Regression**: the application of Robinson's residual-on-residual estimator within a neighborhood of x defined by the forest weights $\alpha_i(x)$. *Extends Robinson's constant-τ estimator to a heterogeneous $\tau(x)$.*

Common Shortnames

Shortname	Meaning
CATE	Conditional Average Treatment Effect
ATE	Average Treatment Effect
HTE	Heterogeneous Treatment Effects
SUTVA	Stable Unit Treatment Value Assumption
OVB	Omitted Variable Bias
OLS	Ordinary Least Squares
S-/T-/X-Learner	Single / Two-model / Cross meta-learner strategies
$e(x)$	Propensity score $\mathbb{P}(W = 1 \mid X = x)$
$m(x)$	Marginal outcome mean $\mathbb{E}[Y \mid X = x]$
$\mu^{(w)}(x)$	Arm-specific outcome regression $\mathbb{E}[Y \mid X = x, W = w]$
$\alpha_i(x)$	Forest adaptive weights